

Kshitiz Neupane

Hattiesburg, MS | kneupane999@gmail.com | 601-913-8554 | LinkedIn | github.com/kshitiz-tech

EDUCATION

University of Southern Mississippi

B.S. Computer Science & Mathematics GPA: 4.0 / 4.0

Hattiesburg, MS

Expected May 2028

- Relevant Coursework: Probability & Statistics, Linear Algebra, Differential Equations, Discrete Mathematics

RESEARCH EXPERIENCE

Research Assistant — ML, NLP & Cybersecurity

Aug 2024 – Present

University of Southern Mississippi

Hattiesburg, MS

- Developed end-to-end data pipelines integrating NLP feature extraction with Graph Neural Network (GNN) architectures for risk and anomaly detection in organizational communication datasets.
- Performed rigorous statistical analysis on text, metadata, and network-structured data; engineered contextual embeddings and linguistic features to surface actionable behavioral patterns.
- Designed anchor-based comparison methods and maintained systematic experiment logs documenting model performance vs. baselines, driving data-informed hypothesis iteration.
- Produced clear diagnostic reports and visualizations communicating failure modes and empirical findings to multi-functional stakeholders.

PROJECTS

Siloed Exact Unlearning (SEU) | Python, PyTorch, HuggingFace Transformers, BERT

GitHub

- Co-authored a machine unlearning method achieving **bitwise-exact, sub-millisecond data deletion** from transformer NLP models; evaluated on IMDb, AG News, and DBpedia with post-unlearn accuracy within 0.3% of retrain oracle.
- Designed N quarantined bottleneck adapter silos atop a frozen BERT backbone; unlearning reduces to re-initialising one adapter to noise — no gradient computation, $O(1)$ latency (6.7 ms vs. ~ 2870 ms retrain oracle, $\sim 428\times$ faster than SISA).
- Validated privacy via membership-inference-attack AUC collapsing to ≈ 0.49 post-unlearn, matching a never-trained control; proved zero cross-silo information leakage via mutual information argument.

WorkForge AI | Agentic RAG Project Management Platform | React, FastAPI, Node.js/Express, ChromaDB, sentence-transformers, MCP

- Built an agentic AI project-management platform that reviews deliverables against project requirements using a local RAG pipeline with ChromaDB and sentence-transformers/all-MiniLM-L6-v2.
- Implemented multi-format document ingestion for PDF, DOCX, Markdown, and TXT files, including chunking, metadata tracking, persistent vector storage, and semantic search retrieval.
- Developed a 5-step deliverable review agent that retrieves deliverable data, queries relevant project context, analyzes compliance, generates improvement suggestions, and compiles review reports.
- Shipped a full-stack React/Vite frontend and FastAPI backend covering search, review workflows, approvals, and PDF proposal generation.

F1 Race Analysis & Predictive Modeling | Python, Pandas, Scikit-Learn, StatsModels, Seaborn

GitHub

- Built end-to-end ML pipelines on 60+ years of historical F1 data: podium classification (ROC-AUC 0.943, Accuracy 88.7%) and finish-position regression (MAE 3.554), with thorough EDA and interactive dashboards.
- Engineered pre-race features (qualifying delta, grid position, rolling driver/constructor form); enforced strict causal separation to ensure valid real-world deployment; packaged reproducible `joblib` inference artifacts.

TECHNICAL SKILLS

Data Analysis & Statistics: Python (Pandas, NumPy, SciPy, Scikit-Learn, StatsModels), R, Statistical Modeling, Hypothesis Testing, Regression, Classification, A/B Analysis

Visualization & Reporting: Matplotlib, Seaborn, Plotly, Dashboard Design, Exploratory Data Analysis

Machine Learning: Bayesian Inference, Supervised Learning, NLP, GNN, LoRA Fine-tuning, Model Evaluation

Systems & Dev: SQL, Git, Linux (Pop!_OS), REST APIs, Django, React/TypeScript

Foundations: Linear Algebra, Probability Theory, Optimization, Differential Equations